The Anisotropy and Polarization of Stress-Induced Microwave Radiation Changes in Granite and Their Significance for Seismic Remote Sensing

Xiang Gao, Shanjun Liu, *Member, IEEE*, Lixin Wu, Wenwen Li, Jianwei Huang, *Member, IEEE*, and Wenfei Mao

***Abstract*— Incidence angles (θ) and polarization modes (p) are the essential factors for detecting earthquake (EQ)-related microwave radiation anomalies. However, their influences are still not well studied. In this article, research was carried out to simulate the observation of satellite on EQ under different θ and p. The microwave observation experiments were performed on granite samples pressed with θ from 0° to 30°, as well as vertical (v) to horizontal (h) polarizations under weak radiation environment. It was found that the stress-induced changes (ΔT_B) in the samples' microwave brightness temperature (T_B) are anisotropic and polarized. It was observed that the trends of ΔT_B may be reversed under different θ (p), with a probability of 57.14% (14.29%). The magnitude of ΔT_B at h is also prone to greater changes than that at v, with a probability of 67.86%. Theoretical analysis was performed based on crystal optics, Fresnel formulas, and dielectric physics. This phenomenon may relate to the combined action of piezoelectric effect, extension and contraction of ionic bonds, and turning-direction polarization. The sensitivity of T_B to the change in permittivity was analyzed by theoretical simulation in order to reveal the effects of modification in θ and p on the T_B change in stress of rock. For rocks with permittivity $\varepsilon = 7 - i0.3$, the most and least sensitive θ are near 52° at h polarization and 69° at v polarization, respectively. Finally, the significance of these results was discussed.**

***Index Terms*— Anisotropy, granite, microwave radiation, polarization, stress, weak background radiation.**

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I. Introduction

EARTHQUAKE (EQ) is a major natural disaster that endangers human life and causes significant economic loss. Remote sensing techniques have been increasingly used for EQ research since the 1970s, with satellite images first being available [1]. Currently, interferometric synthetic aperture radar (InSAR), global positioning system (GPS), infrared remote sensing, and other remote sensing techniques have been widely employed in the detection of EQ-related anomalies [2]–[7].

The various advantages of microwave, such as strong penetration ability and being less affected by cloud in comparison with infrared radiation, have attracted the researchers' attention in utilizing the microwave anomalies in EQ research. In 2009, for the first time, microwave anomalies associated with Al Hoceima EQ (on February 24, 2004 in Morocco) were reported [8]. After that, several efforts have been made to study EQ-related microwave anomalies throughout the world [9]–[17]. For example, microwave anomalies associated with Yushu EQ (on April 12, 2010 in China) were analyzed based on the microwave brightness temperature (T_B) detected by AMER-E under vertical (v) and horizontal (h) polarizations at 18.7 and 23.8 GHz [18]. Wenchuan EQ (on May 12, 2008 in China) was investigated by h polarization T_B of AMER-E at 18.7 GHz [19]. Manyi EQ in 1997 and Kokoxili EQ in 2001 were investigated based on h and v polarizations, with the T_B detected by SSM/I on DMSP satellite at 19.35, 23.8, 37.0, and 85.5 GHz, respectively [20]. Aqua and DMSP are the meteorological satellites that aim at detecting surface and atmospheric temperatures. The microwave radiometers (MWRs) mounted on these satellites can be utilized to calculate T_B for the EQ studies. There are no satellites specifically designed for the detection of EQ-related microwave anomalies, and whether the MWRs on meteorological satellites are suitable is unknown. Besides Aqua and DMSP, there are other satellites such as SMAP, GPM, GCOM-W1, TRMM, and Coriolis that carry MWRs, whose incidence angles are mostly in the range of 40°–60° with h and v polarizations. It is important to ascertain the best use of microwave data from these satellites and to determine which of them could be more suitable for

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studying EQs. Furthermore, it is also crucial to coordinate multisatellite T_B to extract anomalies in microwave signals that are associated with EQs. Therefore, it is essential to study the microwave response of stressed rocks under different incidence angles and h and v polarizations.

Imitative experimental studies based on the microwave observation on loaded rocks in laboratory conditions have been conducted since the 1990s, and some important findings have been achieved [21]–[27]. For example, the microwave observation experiments were carried out on pressed rock [24] and concrete [25] samples by using MWRs with different wavelengths under h and v polarizations. The results of these experiments show that with the increasing of loading, the changes of h polarization T_B in most rock samples are larger than changes of v polarization T_B at 2- and 10-cm wavelengths, while at 3-cm wavelength, the changes of v polarization T_B are more obvious, which illustrates it is important for choosing the best polarization mode in different wavelengths [26]. These studies considered the influence of polarization modes. However, the effects of incidence angles under different polarization modes were not involved. In addition, previous laboratory studies focused mostly on studying microwave emission from rocks loaded in indoor conditions. In recent years, the influence of background radiation has been a cause of concern, due to the fact that indoor background radiation with multiple reflections is much stronger than that of outdoor [28]. In order to better simulate the background radiation of satellite-based observation on the Earth, the microwave observation experiments of rock loaded under 0° incidence angle were conducted in outdoor with weak sky-background radiation [28]–[30]. It was found that under weak background radiation, T_B of rock varies significantly with stress [28]. Moreover, the experiments in 0° incidence angle demonstrated three changing modes of T_B in granite samples with the load increasing, namely, down, up, and no-trend fluctuation patterns [29]. However, up to now, the impact of incidence angle and polarization modes on the stress-induced T_B changes under weak background radiation remains unclear.

In this study, granite was used as the rock sample and T_B of loaded rock samples was observed under different incidence angles and polarization modes in outdoor with weak sky-background radiation. Moreover, the experimental phenomena and their significance for seismic remote sensing were also discussed. The results are helpful in detecting, extracting, and understanding the microwave radiation anomalies associated with EQs.

II. Methodology

A. Experimental Apparatus

A load testing machine and an MWR (shown in Fig. 1) were used in the experiments. The load testing machine has four wheels that can be easily moved from indoor to the outdoor experiment site. It can provide 5000-kN pressure for a sample with a maximum size at 350 × 350 × 600 mm^3. A C-band MWR was used to observe T_B of rock specimens. The MWR has a rectangular horn antenna that is linearly polarized with a 3-dB beam angle of 15°. The center frequency is 6.6 GHz, the middle frequency bandwidth is 400 MHz, and the sensitivity is 0.1 K. The antenna aperture is rectangular with a size of 15 cm × 20 cm. The short side is parallel to the electric field direction of the received microwave. The sampling frequency is 2 Hz in the tests. For a more detailed introduction about the experimental apparatus, readers can refer to relevant articles [28]–[30].



Fig. 1. Showcase of the outdoor experimental system.

B. Experimental Conditions

Granite being readily available on the Earth's surface is selected to make samples. After certification test, the samples are mainly composed of 45% plagioclase, 35% quartz, and 15% potassium feldspar. Seven samples with a size of 600 × 250 × 150 mm^3 were produced and labeled from G-1 to G-7. The nonparallelism of the samples is within 0.5 mm.

Similar to the experimental settings reported in [29], the rock samples were loaded uniaxially at its elastic deformation stage with cyclic mode. According to rock mechanics, the deformation of rock is recoverable at the elastic stage; thus, it is reasonable to observe T_B of the same sample under different incidence angles and different polarization modes. In the loading process, the stress control method was utilized. For the cyclic loading, the initial load was 100 kN (or 2.67 MPa), the maximum load was 1200 kN (or 32 MPa), and loading or unloading rate was 10 kN/s. When the load reached up to 1200 kN or went down to 100 kN, the stress was kept unchanged for 5 s.

Fig. 2 demonstrates the schematic of the placement of MWR. In Fig. 2, a point "O," at which the extension line of the axis in the MWR intersects the upper surface of the specimens, is the center of the observation area of MWR. To ensure the center of the observation area of MWR always be the same when incidence angles and polarization modes are changed, the most of observation area of MWR under different incidence angles and polarization modes should be coincident in the experiments. According to this principle, four elements need to be considered.

1) Observation Position: The observation position determines the areas on the samples that are to be observed. If the

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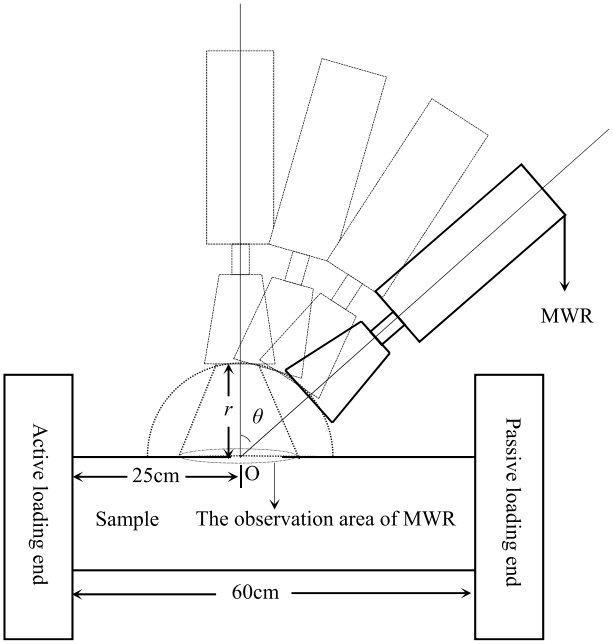


Fig. 2. Schematic of the placement of MWR. In this figure, O is the center of the observation area of MWR. r refers to the distance from point O to the center of antenna aperture. θ refers to incidence angle that is the angle between the MWR axis and the direction perpendicular to the observation surface.

observation position is too close to the left loading end, the observed T_B may be affected by the presser. On the other hand, if the observation point is too far away from the left loading end, the scope of incidence angle will be restricted by the right vertical bar of the presser. Considering above influences, one observation area, whose center is 25 cm away from the active loading end of the presser, was selected for the experiments.

2) Turning Radius (r): The turning radius refers to the distance from point O to the center of antenna aperture. The turning radius is the same under different incidence angles. Since the observation area of MWR increases as r increases, r should not be too large. Otherwise, the observation area of MWR may be far bigger than the size of the samples, and it is difficult to guarantee that the main signal detected by the MWR comes from the sample. If r is too small, the antenna may touch sample surface with the increase of incidence angle and the scope of incidence angles will be restricted. Therefore, r was chosen as 6 cm, at which it can meet the requirement of incidence angles from 0° to 30°.

3) Incidence Angle (θ): The incidence angle refers to the angle between the MWR axis and the direction perpendicular to the observation surface. Under ideal conditions, for the scope of θ, the bigger value is considered better. However, due to the maximum sample length of 600 mm that is allowed by the press, if θ's value gets too high, the detected T_B value may be affected by the loader and the results will be influenced. Therefore, in our experiments, four incidence angles 0°, 10°, 20°, and 30° were selected, respectively. It is also worth noting that the observation experiments under incidence angles from 30° to 90° were not conducted due to limitations of the experimental devices.

4) Polarization Mode (p): In this article, two polarization modes of h and v were considered. The h polarization refers to the direction of the electric field parallel to the sample surface. The h polarization can be converted into v polarization by rotating the antenna 90° around the axis of MWR. When θ is 0°, the antenna's electric field direction is perpendicular (parallel) to the pressure direction under h (v) polarization.

C. Experimental Procedure

First, the load testing machine was moved from indoor to the outdoor experimental field [29] that of weak sky-background radiation. The machine was then adjusted in accordance with the product specification and experimental requirements.

Second, place the sample G-1 into the presser's loading chamber. The polytetrafluoroethylene (PTFE) plates were set between the loading end and the sample end. The placement of plates would weaken not only the friction effect of end faces but also the influence of heat conduction from the loading end. In order to avoid the damage induced by stress concentration, the sample was first placed in coincidence with the axis of the loading end, and then the loads were gradually increased to 100 kN and kept unchanged afterward.

Next, the experiment was started under incidence angle of 0° and h polarization. The MWR was fixed above the rock sample. After turning on the MWR, its time label was aligned to that of the loading machine. When the load was increasing, the T_B value was synchronously recorded. After two times of cyclic load, an observation was over. The above steps were then repeated after the incidence angle was switched by another from 0° to 30° in the step of 10°. After this, the polarization mode was changed to v polarization, and the above steps were conducted again.

Finally, these experiments were performed on each of the seven samples. After the loading experiments were over, 56 observations of the seven samples, each with four incidence angles and two polarization modes, were conducted by the MWR.

III. Experimental Results

A. Stress-Induced T_B Changes (ΔT_B) of the Samples

In order to better analyze the experimental results, the 56 observations were numbered. The observation No. consists of three parts, sample No. (G-i, $i = 1, 2, \ldots, 7$), polarization mode (h or v), and incidence angle (0°, 10°, 20°, or 30°). For example, the observation No. of G-3-h-30 is made up of G-3 (sample No.), h (polarization mode), and 30 (incidence angle). The other observations are numbered similarly. The experimental results show that when the incidence angles and polarization modes changed, variations of T_B were diverse; it could increase, decrease, or fluctuate as the loading stress changes. Fig. 3 shows the representative variation curves of T_B during the cyclic loading process.

In Fig. 3(a), the incidence angle is 30°, and T_B of the observation G-3-h-30 rises (falls) with the stress increasing (decreasing). The correlation coefficient of 0.8345 reflects a significant positive correlation between T_B and stress. In Fig. 3(b), the incidence angle is 10°, and when the stress

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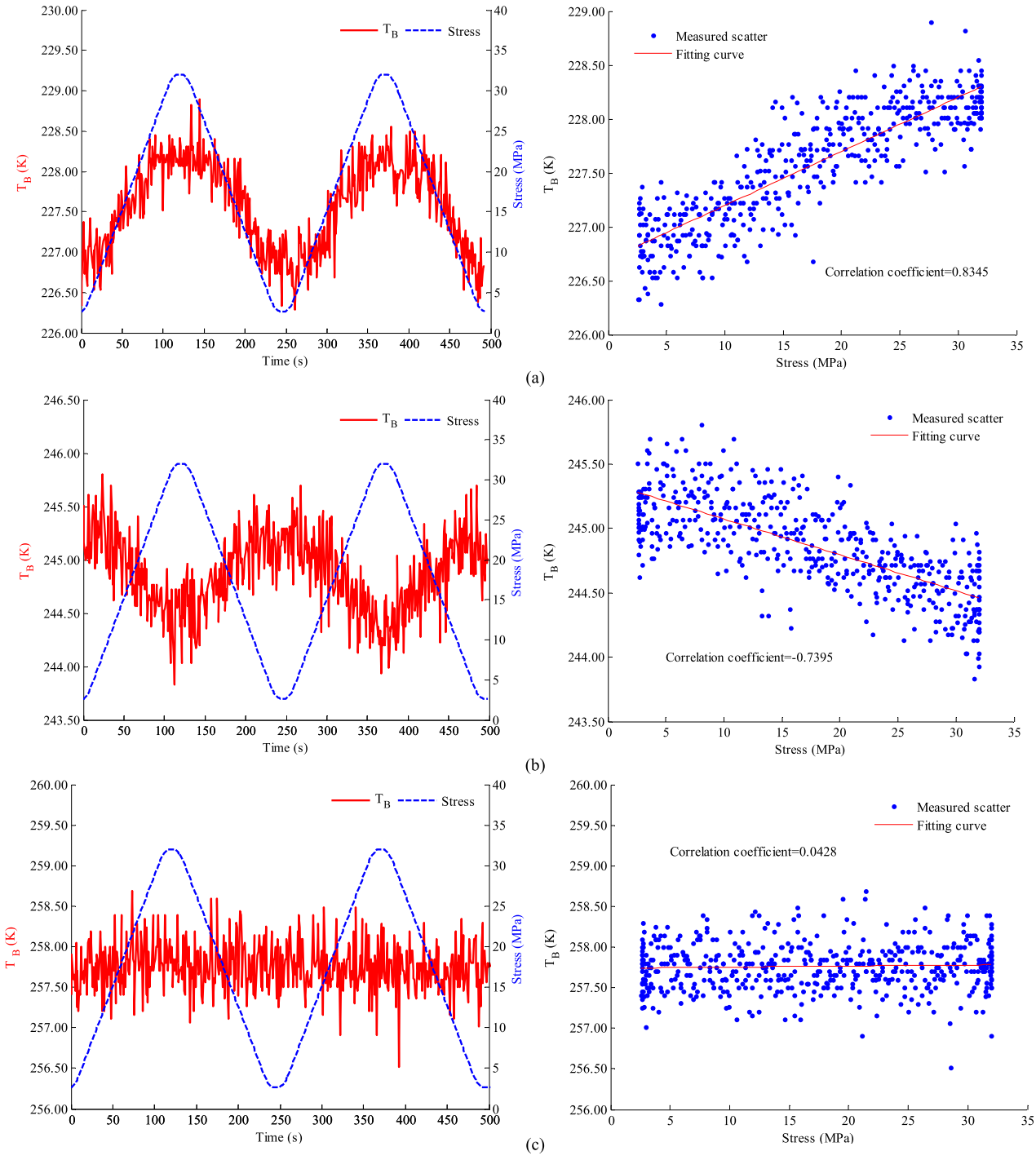


Fig. 3. Representative change of granite T_B during loading process. (a) Curves of the observation No. of G-3-h-30. (b) Curves of the observation No. of G-3-h-10. (c) Curves of the observation No. of G-2-v-10.

increases (decreases), T_B of G-3-h-10 falls (rises). The correlation coefficient of -0.7395 reflects a significant negative correlation. In Fig. 3(c), the incidence angle is also 10°, but the polarization has changed into v polarization, and it shows that T_B of G-2-v-10 fluctuates with the change of stress. The stress-induced variations of T_B (ΔT_B) of all samples are listed in Table I, which illustrates the numbers of samples with uptrend, downtrend, and fluctuation types of stress-induced T_B changes are 17, 19, and 20, respectively, with corresponding probabilities of 30.36%, 33.93%, and 35.71%.

Due to the linear correlation between T_B and stress, the equation of ΔT_B can be written as follows:

$$\Delta T_B = \hat{a} \cdot \Delta\sigma \tag{1}$$

where $\hat{a}$, determined by the least-squares method, is the rate at which T_B changes with stress; $\Delta\sigma$ is the change of stress. Here, the trends (magnitude) of ΔT_B refers to the trends (magnitude) of stress-induced T_B change with $\Delta\sigma$, where $\Delta\sigma = 29.33$ MPa in the experiment.

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TABLE I

STRESS-INDUCED CHANGES IN T_B OF THE SAMPLES UNDER DIFFERENT INCIDENCES AND POLARIZATIONS

Sample No.	Incidence angle (°)	*h*			*v*		
		R	ΔT_B (K)	Correlation	R	ΔT_B (K)	Correlation
G-1	0	-0.0421	-0.04	Irrelevant	-0.2137	-0.21	Negative
	10	-0.0893	-0.09	Irrelevant	0.1224	0.12	irrelevant
	20	0.2155	0.25	Positive	0.2431	0.58	Positive
	30	0.0454	0.06	Irrelevant	0.3241	0.34	Positive
G-2	0	-0.6892	-0.97	Negative	-0.0958	-0.07	Irrelevant
	10	-0.6476	-0.79	Negative	0.0428	0.04	Irrelevant
	20	-0.8330	-1.37	Negative	-0.2014	-0.45	Negative
	30	-0.7900	-1.18	Negative	0.0874	0.07	Irrelevant
G-3	0	-0.4581	-0.41	Negative	-0.3724	-0.34	Negative
	10	-0.7395	-0.82	Negative	0.4825	0.46	Positive
	20	0.6350	0.77	Positive	0.5474	0.58	Positive
	30	0.8345	1.41	Positive	0.4898	0.49	Positive
G-4	0	-0.0313	-0.03	Irrelevant	-0.4862	-0.61	Negative
	10	-0.2948	-0.28	Negative	0.5281	0.52	Positive
	20	0.5150	0.53	Positive	0.2093	0.15	Irrelevant
	30	0.8119	1.30	Positive	0.6165	0.71	Positive
G-5	0	-0.7201	-1.14	Negative	-0.0530	-0.05	Irrelevant
	10	-0.7429	-1.00	Negative	0.2108	0.19	Positive
	20	-0.4612	-0.42	Negative	-0.1086	-0.10	Irrelevant
	30	0.7789	1.07	Positive	-0.3606	-0.32	Negative
G-6	0	-0.0017	0.00	Irrelevant	0.1061	0.10	Irrelevant
	10	0.0098	0.01	Irrelevant	0.4381	0.42	Positive
	20	-0.4328	-0.49	Negative	0.1628	0.14	Irrelevant
	30	-0.5262	-0.54	Negative	0.1017	0.08	Irrelevant
G-7	0	0.6332	0.71	Positive	0.0104	0.01	Irrelevant
	10	0.2329	0.22	Positive	0.0223	0.02	Irrelevant
	20	0.2591	0.24	Positive	-0.1179	-0.10	Irrelevant
	30	-0.1120	-0.10	Irrelevant	0.2723	0.25	Positive

Table Note: R refers to the correlation coefficient of T_B and stress. The magnitude and trends of ΔT_B is recorded when the stress is increased by 29.33 MPa.

B. ΔT_B of the Samples Under Different Incidence Angles and Polarization Modes

In the experiment, three variables are involved, including incidence angles, polarization modes, and sample No., and three features of ΔT_B were illustrated: anisotropy, polarization, and nonuniformity.

1) Anisotropy of the Samples' ΔT_B: Fig. 4 shows the ΔT_B-incidence angles curves of the samples at h polarization. The analysis of Fig. 4 and Table I shows that ΔT_B of each sample varies with the change of incidence angles, but there is no uniform law to follow. For some samples, e.g., G-5, ΔT_B increases with the increase of incidence angles, while ΔT_B of other samples, e.g., G-7, decreases. This suggests that ΔT_B of the samples is anisotropic. It is worth noting that not only the magnitude of ΔT_B varies with the change of incidence angles but also the trends of ΔT_B may be opposite at different incidence angles, with a probability of four of seven (or 57.14%). For example, the ΔT_B symbol of G-5 under 10° incidence angle is different from that under 30°, and it is similar for the samples of G-3, G-4, and G-7. In addition, ΔT_B at v polarization is similar to that at h polarization and it would not be repeated. The phenomenon that the trends of ΔT_B are opposite under different incidence angles may serve as an essential foundation for seismic remote sensing research. This point will be further discussed below.

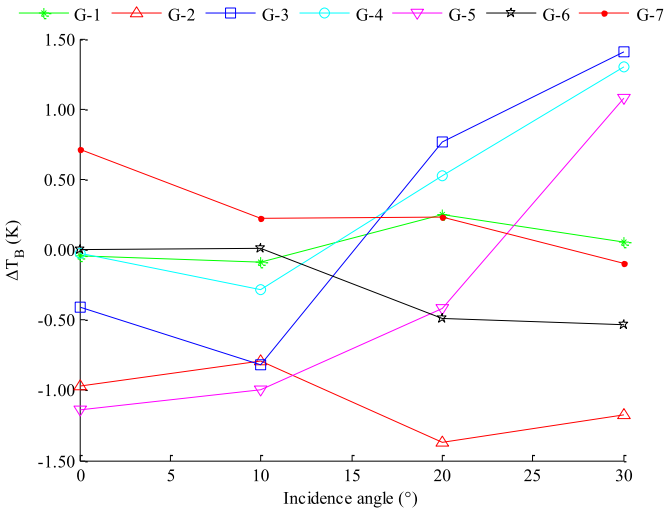


Fig. 4. ΔT_B—incidence angles curves of the specimens at *h* polarization with the stress increasing by 29.33 MPa. The seven curves with different colors and point types correspond to the change of ΔT_B of the seven samples at different incident angles.

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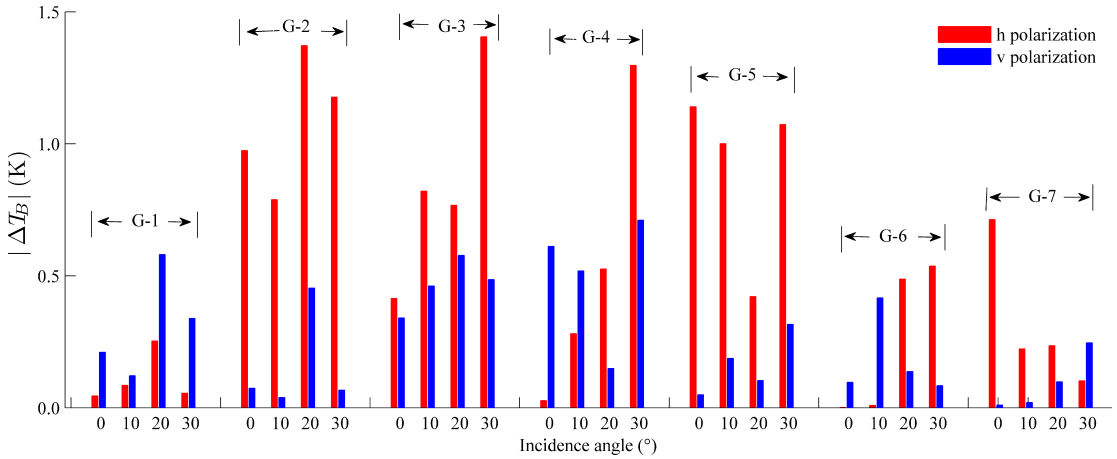


Fig. 5. Magnitude of ΔT_{B} ($|\Delta T_{\mathrm{B}}|$) of the samples at h and v polarizations. In this figure, G-1 refers to sample No., and so are the others from G-2 to G-7; 0, 10, 20, and 30 who are below the abscissa refer to the incidence angles.

TABLE II
STATISTICS ON THE NONUNIFORMITY OF STRESS-INDUCED VARIATION TRENDS IN T_{B} BETWEEN V AND H POLARIZATIONS

Relationship	Number	Possibility (%)
Opposite	4	14.29
Same	6	21.43
Other	18	64.28
Summary	28	100.00

Table Note: "Opposite" refers to that the trend of ΔT_B at h is uptrend (or downtrend), and, at the same time, that at v is downtrend (or uptrend); "Same" refers to that the trends of ΔT_B at h *and* v are same, both uptrend, downtrend or fluctuation; "Other" refers to that the trend of ΔT_B at h is uptrend or downtrend (fluctuation), and, at the same time, that at v is fluctuation (uptrend or downtrend).

2) Polarization of the Samples' ΔT_{B}: Table I shows that ΔT_{B} varies with polarization modes under the same incidence angle, which suggests that ΔT_{B} of the samples is polarized. The polarization is presented in two aspects, the magnitude and the trends of ΔT_{B}.

Fig. 5 shows the magnitude of ΔT_{B} ($|\Delta T_{\mathrm{B}}|$) of the samples under h and v polarization modes. From Fig. 5, it can be seen that $|\Delta T_{\mathrm{B}}|$ at h polarization modes is higher than that at v, with a probability of 19 in 28 (67.86%). The computational analysis shows that the average value of $|\Delta T_{\mathrm{B}}|$ at h polarization is 0.58 K, and that at v is 0.27 K. Therefore, the former (0.58 K) is also greater than the latter (0.27 K), which suggests that $|\Delta T_{\mathrm{B}}|$ of the samples at h polarization modes is higher than that at v.

Table II shows the statistics on the nonuniformity of stress-induced variation trends of T_{B} between h and v polarizations, which demonstrates that 21.43% samples have similar trends of ΔT_{B} between h and v polarizations, while 14.29% have opposite trends. For example, Table I clearly shows that when G-3 is subjected to 10° incidence angle, ΔT_{B} at h is −0.82 K, while that is 0.46 K at v polarization. The phenomenon that ΔT_{B} shows an opposite trend under different polarization modes may also play an essential role in seismic remote sensing research. It will be discussed below.

3) Nonuniformity of the Samples' ΔT_{B}: The experiments in [29] illustrated different trends of ΔT_{B}, including uptrend, downtrend, and fluctuation, are nonuniform under 0° incidence angle for granite samples. A similar phenomenon occurred in this study. In addition, ΔT_{B} of the samples in other incident angles is also not uniform. For example, T_{B} in the observation of G-3-h-20 increases with the increase of stress, while the observed T_{B} of G-2-h-20 decreases with the increase of stress. Therefore, the nonuniformity of ΔT_{B} in different incidence angles demonstrates a certain universality for granite samples.

C. Average Magnitude of ΔT_{B} ($\overline{|\Delta T_{\mathrm{B}}|}$) of Granite Under Different Incidence Angles and Polarization Modes

The variations in T_{B} of the samples are described above. However, the changes of ΔT_{B} with incidence angles are different with individual samples, and each granite sample cannot represent the full spectrum of granites. It was found that the nonuniformity of ΔT_{B} in magnitude and symbols was caused by the inhomogeneity of different granite samples [29]. Here, a data processing method was used to weaken the effect of the inhomogeneity in granite samples, by which ΔT_{B} of the samples was first changed into absolute value and was averaged later. The processed ΔT_{B} is written as follows:

$$\overline{|\Delta T_{\mathrm{B}}|} = \frac{1}{n}\sum_{i=1}^{n} |\Delta T_{Bi}| \tag{2}$$

where $\overline{|\Delta T_{\mathrm{B}}|}$ reflects the average magnitude of ΔT_{B} in granite; n refers to the number of samples; ΔT_{Bi} represents the ΔT_{B}

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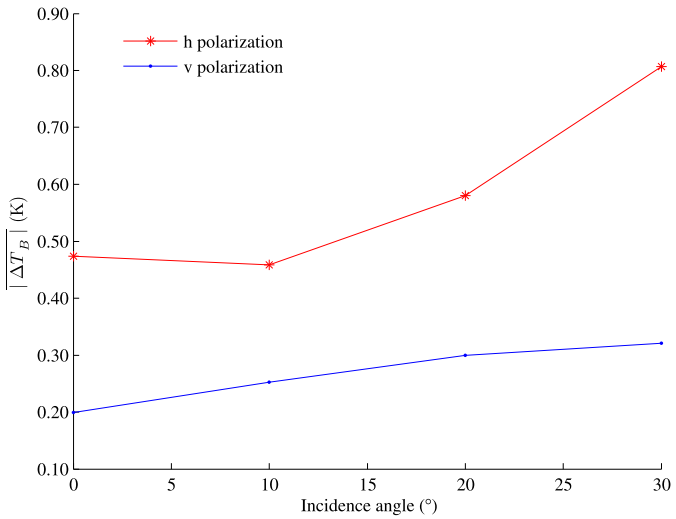


Fig. 6. $\overline{|\Delta T_B|}$-incidence angle curves of granite at h and v polarizations.

of the ith sample under the same incident angle and polarization mode. Hereby, the general ΔT_B-incidence angle law of granite was extracted. The $\overline{|\Delta T_B|}$-incidence angle curves under h and v polarizations are shown in Fig. 6.

It can be deduced from Fig. 6 that $\overline{|\Delta T_B|}$ of granite increases with the increase of incidence angles 0°–30° under h and v polarizations. $\overline{|\Delta T_B|}$ under h polarization at each incidence angle is always greater than that under v, which demonstrates that $\overline{|\Delta T_B|}$ of granite is polarized, and the response of T_B is more sensitive to the change of rocks' stress state under h polarization. $\overline{|\Delta T_B|}$ reaches the maximum value at h polarization and 30° incidence angle, which is the optimal observing condition when the incidence angle is between 0° and 30°.

IV. Theoretical Simulation of $\overline{|\Delta T_B|}$ Based on Fresnel Formulas

In this experiment, the incidence angle is confined to the range of 0°–30° due to the limitation of the experimental devices. Moreover, our preliminary experiments show that the most sensitive incidence angles and polarization modes are 30° and h polarization, respectively. If we consider the sensitive incidence angles and polarization modes with incidence angle in the range of 0°–90°, a theoretical simulation method may be available to make up for the limitation of the experimental devices. However, an important prerequisite for using the theoretical simulation method is to see whether the results of experiments and theoretical simulation with incidence angle from 0° to 30° are consistent.

A. Relationship Between ΔT_B and Incidence Angles and Polarizations

When the rock is stressed, its physical properties may change. The stress-induced changes (ΔT_B) in T_B are as follows [29]:

$$\Delta T_B = e\Delta T_{obj} + \Delta e(T_{obj} - T_{sur}) \tag{3}$$

where e is rock's emissivity; Δe is the change of emissivity; T_{obj} is object's physical temperature; T_{sur} is environmental T_B. In (3), $e\Delta T_{obj}$ and $\Delta e(T_{obj} - T_{sur})$ are the contribution from change of rock's temperature and emissivity, respectively. Based on [29], the contribution of $e\Delta T_{obj}$ is minimal during the short experiment period with relative stable temperature state at night time, and it can be ignored. Therefore, ΔT_B is approximately equal to

$$\Delta T_B = \Delta e(T_{obj} - T_{sur}). \tag{4}$$

According to Fresnel formulas [32] and the law of conservation of energy, the relationship between emissivity at h and v polarizations and dielectric constant (ε) can be gained as follows:

$$e_h(\varepsilon,\theta) = 1 - \left|\frac{\sqrt{\varepsilon - \sin^2(\theta)} - \cos(\theta)}{\sqrt{\varepsilon - \sin^2(\theta)} + \cos(\theta)}\right|^2 \tag{5}$$

$$e_v(\varepsilon,\theta) = 1 - \left|\frac{\sqrt{\varepsilon - \sin^2(\theta)} - \varepsilon\cdot\cos(\theta)}{\sqrt{\varepsilon - \sin^2(\theta)} + \varepsilon\cdot\cos(\theta)}\right|^2 \tag{6}$$

where $\varepsilon = \varepsilon' + i\varepsilon''$ is the complex dielectric permittivity; θ is the incidence angle. Given the variation of ε is $\Delta\varepsilon$, combining (5) and (6), Δe can be written as follows:

$$\Delta e_h(\varepsilon,\theta,\Delta\varepsilon) = e_h(\varepsilon+\Delta\varepsilon,\theta) - e_h(\varepsilon,\theta) \tag{7}$$

$$\Delta e_v(\varepsilon,\theta,\Delta\varepsilon) = e_v(\varepsilon+\Delta\varepsilon,\theta) - e_v(\varepsilon,\theta). \tag{8}$$

Combining (7), (8), and (4), ΔT_B at h and v polarizations can be transformed into

$$\Delta T_{Bh} = \Delta e_h(\varepsilon,\theta,\Delta\varepsilon)\cdot(T_{obj} - e_{sur}T_{sur}) \tag{9}$$

$$\Delta T_{Bv} = \Delta e_v(\varepsilon,\theta,\Delta\varepsilon)\cdot(T_{obj} - e_{sur}T_{sur}). \tag{10}$$

B. Theoretical Simulation of the Anisotropy and Polarization of $\overline{|\Delta T_B|}$

The changes of dielectric constant are the main reason for the stress-induced T_B changes of rocks. $\overline{|\Delta T_B|}$ under different incidence angles and polarization modes can be simulated based on (9) and (10). In (9) and (10), five parameters, ε', ε'', $\Delta\varepsilon$, T_{Bsur}, and T_{obj}, need to be determined if θ is considered as an independent variable. As for the microwave dielectric constant of granite at 6.9 GHz, ε' is within the range of [3.45, 5.97], and ε'' value is between 0.02 and 0.387 [33]. Thus, the average values for $\varepsilon' = 4.70$ and $\varepsilon'' = 0.15$ were taken for the granite samples (see Table III). Previous studies have found that the imaginary part of permittivity of feldspar specimen at 6 GHz does not change significantly with stress [34]; thus, we assume that ε'' of granite specimens in the experiments changes slightly. Ignoring the differences of the background radiation under different incidence angles, T_{sur} and T_{obj} can be taken as the sky T_B of 15 K and the environment temperature of 20 °C, i.e., 293.15 K, respectively. When there is a 29.33-MPa increase in stress, the quartz permittivity will decrease by 0.08 [29]. Given that the stress-induced permittivity increment taken as −0.08 and it is the same in each direction, $\overline{|\Delta T_B|}$ simulated under the incidence angles within the range of 0°–30° is shown in Fig. 7.

In Fig. 7, $\overline{|\Delta T_B|}$ under v (h) polarizations decreases (increases) with an increasing incidence angle.

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TABLE III
VALUES OF THE FIVE PARAMETERS IN (9) AND (10)

ε'	ε''	$\Delta\varepsilon'$	T_{obj} (K)	T_{sur} (K)
4.70	0.15	-0.08	293.15	15

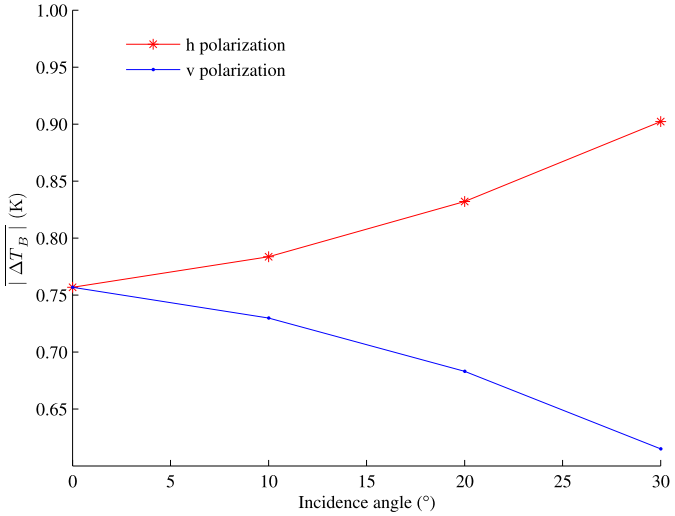


Fig. 7. $\overline{|\Delta T_B|}$-incidence angle curves at h and v polarizations by theoretical simulation.

Comparing Fig. 7 with Fig. 6, it can be concluded that the change trends of $\overline{|\Delta T_B|}$ with incidence angles under v (h) polarizations by simulation method are different from (consistent with) that by the experimental method.

V. DISCUSSION

A. Explanation of the Experimental Phenomena

1) Reasons for the Nonuniformity of ΔT_B: The nonuniformity of ΔT_B under 0° incidence angle is discussed based on the experiments in [29]. In [29], the permittivity of single-crystal quartz, one of the major minerals in granite, was measured with stress loaded in two directions: perpendicular and parallel to the optical axis. It was found that the real part of the permittivity of quartz changes as the stress changes, and the change patterns are associated with the relationship between the directions of stress and the optical axis. When the two axes are perpendicular to each other, they are positively correlated. When the two axes are in parallel to each other, the real part of the permittivity is in negative correlation of the stress. It was considered that the changes of permittivity caused the T_B variation of the pressed granite samples when the background radiation is weak. There is the inhomogeneity of granite samples, namely, random distribution of dielectric constant of minerals inside the granite samples. This results in random variation of minerals permittivity and the nonuniformity of ΔT_B under 0° incidence angle. Therefore, it can be assumed that the mechanisms are similar under other incidence angles. However, the micromechanisms involved in the experimental phenomena still require further discussion. In this article, the micromechanisms by which the relationship between directions of stress and optical axis causes the stress-induced changes of permittivity in quartz are proposed as follows.

According to dielectric physics [35], the polarization of dielectric medium can occur under the electric field's action, the degree of which can be characterized by electric polarization intensity (**P**). **P** can be written as follows:

$$\mathbf{P} = N \cdot \mathbf{p} = N \cdot q \cdot \mathbf{L} \tag{11}$$

where N is molecular number per unit volume; **p** is the mean dipole moment per molecular; q is the charge of dipole moment; **L** is the distance between the centers of gravity of positive and negative charges. There is a linear relationship between **P** and the electric field strength (**E**) for most of the dielectric medium and is expressed as follows:

$$\mathbf{P} = \chi_e \cdot \varepsilon_0 \cdot \mathbf{E} \tag{12}$$

where χ_e is electric susceptibility, which is constant if dielectric medium is isotropic, otherwise varies with directions; ε_0 is the permittivity of vacuum. The permittivity of dielectric medium (ε) can be expressed as follows:

$$\varepsilon = (\chi_e + 1) \cdot \varepsilon_0. \tag{13}$$

Combining (11) and (12), (13) can be transformed into

$$\varepsilon = \frac{Nq\mathbf{L}}{\mathbf{E}} + \varepsilon_0. \tag{14}$$

According to (14), ε relates to N and **L** at a microscopic level. The permittivity of rock materials is affected by many polarization mechanisms, and the main mechanisms may be different for different minerals. When the quartz mineral is compressed, the microstructure will change, which may result in changes in permittivity. Three possible microscopic changes and the effects on permittivity are discussed below.

a) Piezoelectric effect: According to the piezoelectric behavior of minerals [36], the piezoelectric effect does not always occur when piezoelectric minerals are subjected to external forces. For example, when the external force's direction is parallel to the optical axis of quartz, the piezoelectric effect does not occur. The extension and contraction of ionic bond length caused by external force are even and, therefore, will not cause ion polarization. However, when the direction of the external force is perpendicular to the optical axis of quartz, the bond length in the lattice would shrink or expand unevenly. The lattice will, therefore, become irregular, resulting in ion polarization. **L** in (14) increases due to ion polarization after the minerals are pressed, and therefore resulting in the increase of ε. The change of ε caused by the piezoelectric effect can be recorded as $\Delta\varepsilon_i$.

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b) *Extension and contraction of ionic bonds:* According to the photo-elasticity principle [37], the homogeneous isotropic transparent material under stress will deform and become an anisotropic body. This means that the stress can cause the anisotropic change of permittivity of solid minerals. After the crystal is pressed, it deviates from the equilibrium state. As a result, the bond length shortens in the stress direction and elongates in the direction perpendicular to the stress. At an equilibrium, the energy is the lowest, whereas the energy increases when it deviates from equilibrium. The exerted mechanical energy is converted into potential energy between ions, including the electron cloud's repulsion potential and the attraction potential of Coulombic force. The increase of potential energy between ions increases the difficulty of ion polarization under the action of electric field. That reduces **L** in (14), thereby decreasing ε of crystals. The change of ε caused by the extension and contraction of ionic bonds can be recorded as $-\Delta\varepsilon_{\mathrm{e}}$.

c) *Turning-direction polarization of dipoles:* When dipoles exist in crystal, under the electric field's action, turning-direction polarization of dipoles will occur. The quartz crystal is pressed with stress directions perpendicular to its optical axis. This leads to ion polarization, and dipoles are created in the crystal due to the piezoelectric effect. The dipoles may turn directions under the electric field's action, which will increase **L** and ε in (14). The change of ε caused by turning-direction polarization of dipoles can be recorded as $\Delta\varepsilon_{\mathrm{t}}$.

When stress is parallel to the optical axis, the piezoelectric effect does not occur as the quartz is pressed. However, the pressure action can lead to the extension and contraction of ionic bonds. $-\Delta\varepsilon_{\mathrm{e}}$ caused by the extension and contraction of ionic bonds in the process of pressing mineral may be the main reason for the decrease of ε in quartz as the stress is increased. When stress is perpendicular to the optical axis, both the piezoelectric effect and the extension and contraction of ionic bonds will occur after the quartz is pressed. Therefore, the turning-direction polarization of dipoles may also occur under the action of the electric field. The combined action of the piezoelectric effect ($\Delta\varepsilon_{\mathrm{i}}$), the extension and contraction of ionic bonds ($-\Delta\varepsilon_{\mathrm{e}}$), and the turning-direction polarization of dipoles ($\Delta\varepsilon_{\mathrm{t}}$) results comprehensively in the change of ε in quartz with the increase of stress.

2) Reasons for the Anisotropy and Polarization of ΔT_{B}*:* Granite is complex because it is the aggregate of quartz and other minerals. And the changes of minerals' permittivity cause the changes in T_{B} of granite. To discuss the causes by which stress-induced T_{B} in the samples varies with incidence angles or polarization modes, as an example, whether the incidence angles and polarizations impact the changes in permittivity of quartz pressed would be analyzed first.

Since minerals' permittivity is anisotropic and polarized [38], [39], the stress-induced changes of permittivity in minerals may also be anisotropic and polarized. According to crystal optics [40], for isotropic media, the relationship between electric field intensity vector (**E**) and electric displacement vector (**D**) can be expressed as follows:

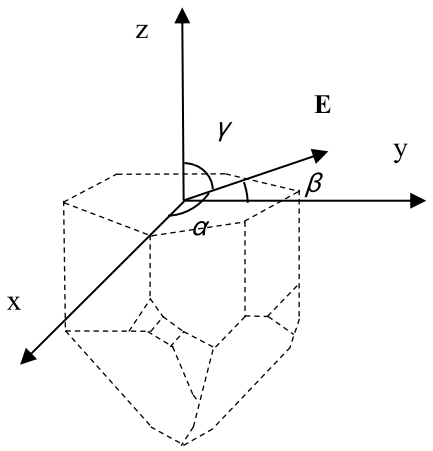


Fig. 8. Schematic of three orthogonal principal axes in quartz.

$$\mathbf{D} = \varepsilon \cdot \mathbf{E}. \tag{15}$$

For anisotropic media, due to the anisotropy of ε, the relationship between **E** and **D** is much complex [41], namely,

$$\mathbf{D} = [\varepsilon] \cdot \mathbf{E} \tag{16}$$

where $[\varepsilon]$ is the second order dielectric tensor of a medium. According to crystal optics [42], the quartz crystal belongs to the hexagonal system and consists of three orthogonal principal axes of the dielectric, as shown in Fig. 8. The dielectric tensor is expressed as follows:

$$[\varepsilon] = \begin{bmatrix} \varepsilon_{xx} & 0 & 0 \\ 0 & \varepsilon_{yy} & 0 \\ 0 & 0 & \varepsilon_{zz} \end{bmatrix} \tag{17}$$

where ε_{xx}, ε_{yy}, and ε_{zz} refer to ε at the direction of x-, y-, and z-axes, respectively. It is noted that ε_{yy} is equal to ε_{xx} for quartz crystal according to crystal optics [42]. According to (16) and (17), under the action of **E** whose direction vector is $(\cos\alpha, \cos\beta, \cos\gamma)$, **D** can be written as follows:

$$\begin{aligned} \begin{bmatrix} D_x \\ D_y \\ D_z \end{bmatrix} &= \begin{bmatrix} \varepsilon_{xx} & 0 & 0 \\ 0 & \varepsilon_{yy} & 0 \\ 0 & 0 & \varepsilon_{zz} \end{bmatrix} \cdot \begin{bmatrix} E_x \\ E_y \\ E_z \end{bmatrix} \\ &= \begin{bmatrix} \varepsilon_{xx} & 0 & 0 \\ 0 & \varepsilon_{yy} & 0 \\ 0 & 0 & \varepsilon_{zz} \end{bmatrix} \cdot \begin{bmatrix} E\cos\alpha \\ E\cos\beta \\ E\cos\gamma \end{bmatrix} = \begin{bmatrix} E\varepsilon_{xx}\cos\alpha \\ E\varepsilon_{yy}\cos\beta \\ E\varepsilon_{zz}\cos\gamma \end{bmatrix} \end{aligned} \tag{18}$$

where α, β, and γ refer to the angles between the direction of **E** with x-, y-, and z-axes, respectively. The vector module of **D** is

$$\begin{aligned} |\mathbf{D}| &= \sqrt{D_x^2 + D_y^2 + D_z^2} \\ &= \sqrt{(\varepsilon_{xx}\cos\alpha)^2 + (\varepsilon_{yy}\cos\beta)^2 + (\varepsilon_{zz}\cos\gamma)^2} \cdot E. \end{aligned} \tag{19}$$

According to the definition of ε and comparing (15) with (19), ε at the direction of $(\cos\alpha, \cos\beta, \cos\gamma)$ can be expressed as follows:

$$\varepsilon_{\langle\cos\alpha,\cos\beta,\cos\gamma\rangle} = \sqrt{(\varepsilon_{xx}\cos\alpha)^2 + (\varepsilon_{yy}\cos\beta)^2 + (\varepsilon_{zz}\cos\gamma)^2}. \tag{20}$$

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$\varepsilon_{\langle\cos\alpha,\cos\beta,\cos\gamma\rangle}$ also is ε_{b} which is ε at the direction $(\cos\alpha, \cos\beta, \cos\gamma)$ before the mineral is pressed. ε at the direction $(\cos\alpha, \cos\beta, \cos\gamma)$ after the mineral is pressed will be discussed below.

As discussed earlier, the changes of ε are related to the angle (φ) between the directions of stress and optical axis. The values of $\Delta\varepsilon_{\mathrm{i}}$, $\Delta\varepsilon_{\mathrm{e}}$, and $\Delta\varepsilon_{\mathrm{t}}$ are all functions of φ. ε at x-, y-, and z-axes directions will change after quartz is pressed. At x-axis, due to the piezoelectric effect and the extension and contraction of ionic bonds, the change of ε is $\Delta\varepsilon_{\mathrm{ix}}(\varphi) - \Delta\varepsilon_{\mathrm{ex}}(\varphi)$. At y- and z-axes, due to turning-direction polarization and the extension and contraction of ionic bonds, the changes of ε are $\Delta\varepsilon_{\mathrm{iy}}(\varphi) - \Delta\varepsilon_{\mathrm{ey}}(\varphi)$ and $\Delta\varepsilon_{\mathrm{tz}}(\varphi) - \Delta\varepsilon_{\mathrm{ez}}(\varphi)$, respectively. Hence, the dielectric tensor of quartz under compression can be expressed as $[\varepsilon]_{\mathrm{a}}$

Parts of (21), as shown at the bottom of the next page, can be simplified as follows:

$$\left.\begin{aligned} -[\Delta\varepsilon_{\mathrm{ix}}(\varphi) - \Delta\varepsilon_{\mathrm{ex}}(\varphi)] &= \Delta F(\varphi) \\ -[\Delta\varepsilon_{\mathrm{iy}}(\varphi) - \Delta\varepsilon_{\mathrm{ey}}(\varphi)] &= \Delta G(\varphi) \\ -[\Delta\varepsilon_{\mathrm{tz}}(\varphi) - \Delta\varepsilon_{\mathrm{ez}}(\varphi)] &= \Delta H(\varphi) \end{aligned}\right\}. \tag{22}$$

Combining (22) with (18), (19), (20), and (21), ε of quartz pressed at the direction of $(\cos\alpha, \cos\beta, \cos\gamma)$ can be gained, (23), as shown at the bottom of the next page.

The difference of ε before and after quartz is pressed at the direction of $(\cos\alpha, \cos\beta, \cos\gamma)$ is written as follows (24), as shown at the bottom of the next page.

As is well known that the polarization mode is related to the directions of the electric field. When the polarization mode varies, the direction of the electric field will also change. This is to say that α, β, and γ are all functions of polarization mode (p). Due to the fact that incident direction of microwave and the electric field direction are perpendicular to each other, the electric field direction may vary with the changes of incident direction. This is to say that α, β, and γ are functions of incident angle (θ), too. Considering $\alpha = f(\theta, p)$, $\beta = g(\theta, p)$, $\gamma = \mathrm{h}(\theta, p)$, (24) can be transferred further into

$$\begin{aligned} \Delta\varepsilon(\varphi, \theta, p) = &\frac{[\Delta F(\varphi)^2 - 2\varepsilon_{xx}\Delta F(\varphi)]\cos^2[f(\theta, p)]}{\varepsilon_{\mathrm{a}} + \varepsilon_{\mathrm{b}}} \\ &+ \frac{[\Delta G(\varphi)^2 - 2\varepsilon_{yy}\Delta G(\varphi)]\cos^2[g(\theta, p)]}{\varepsilon_{\mathrm{a}} + \varepsilon_{\mathrm{b}}} \\ &+ \frac{[\Delta H(\varphi)^2 - 2\varepsilon_{zz}\Delta H(\varphi)]\cos^2[h(\theta, p)]}{\varepsilon_{\mathrm{a}} + \varepsilon_{\mathrm{b}}} \end{aligned} \tag{25}$$

where φ reflects the angle between directions of stress and optical axis, θ refers to incident angles, and p refers to polarization modes. From (25), after quartz is pressed, the changes in ε are associated with φ, θ, and p. The symbols (positive or negative) and size of $\Delta\varepsilon$ are determined by the numerator of (25). Positive or negative $\Delta\varepsilon$ may be appear with the changes of φ, θ, and p. However, more quantitative analysis is still required in the future.

Then, it can be inferred that $\Delta\varepsilon$ in granite samples is also influenced by φ, θ, and p, and the relationship between $\Delta\varepsilon$ of inside minerals and φ, θ, and p is much complex. Due to the fact that the optical axes of the inside mineral crystals are nondirective, φ, θ, and p of each mineral particle are not always the same, thereby resulting in multiple $\Delta\varepsilon$ and ΔT_{B} in different granite samples under the same incident angle and polarization mode or in the same granite sample under different incident angles and polarization modes. Although it has been discussed about the possibility of ΔT_{B} being different under different θ or p theoretically, further quantitative research and experimental verification are still needed in the future.

In addition, based on the Fresnel formulas, even if the change of dielectric constant is isotropic and nonpolarized, the sensibility of T_{B} to the changes of dielectric constant is still a function of θ and p. Therefore, $\overline{|\Delta T_{\mathrm{B}}|}$ of granite is associated with θ and p. More detailed discussion is presented in Section V-B.

3) Cause of the Inconsistency Between the Results of Theoretical Simulation and Experiment: The variation trends of $\overline{|\Delta T_{\mathrm{B}}|}$ in the samples with θ under v polarization by experimental method are different from that of simulation method. This variation perhaps is related to the near-field background radiation in the experiment.

For this experiment, the maximum size of the antenna (D) is 20 cm and the wavelength of center frequency (λ) is 4.5 cm. The maximum distance from the antenna aperture to the observation surface (R) is 6 cm. According to the antenna theory [43], the condition $R < 2D^2/\lambda$ (177.8 cm) belongs to the near field. Besides receiving, the antenna of MWR can also continuously radiate microwave outward, which becomes a part of the background radiation. Under near-field conditions, the microwave emitted by the antenna may be received back again after the target reflects it. With the increase of the incident angle, the microwave radiation, emitted by the antenna and then received by it again, may decrease, thereby causing the background radiation to decrease.

According to (4), there is negative correlation between ΔT_{B} and T_{sur}. If T_{sur} decreases with increasing θ, ΔT_{B} will increase incrementally. To understand the variation of T_{sur} with θ, a microwave-observation experiment on water body with a size of 120 cm × 200 cm × 5 cm (width, length, and depth) under variable incidence angles, outdoor, and near-field conditions was carried out.

Fig. 9 shows the changing curves of T_{B} of the water body with θ under v polarization. In Fig. 9, under v polarization, T_{B} of water decreases with θ in the range of 0°–30°. The background T_{B} was derived with the dielectric constant of water being 81 and θ at 0°–30°. The relationship between background T_{B} and θ is shown in Fig. 9. It can be deduced from Fig. 9 that, under v polarization, the background T_{B} decreases with θ at 0°–30°, which is consistent with the above analysis. In addition, under h polarization, the background T_{B} also decreases with the increase of θ, which is similar to that under v polarization.

The above analysis shows that the background T_{B} (T_{sur}) is a function of θ. According to (7) and (8), Δe is also a function of θ. Therefore, according to (4), $|\Delta T_{\mathrm{B}}|$ can be written as follows:

$$|\Delta T_{\mathrm{B}}(\theta)| = |\Delta e(\theta)| \cdot [T_{\mathrm{obj}} - T_{\mathrm{sur}}(\theta)] \tag{26}$$

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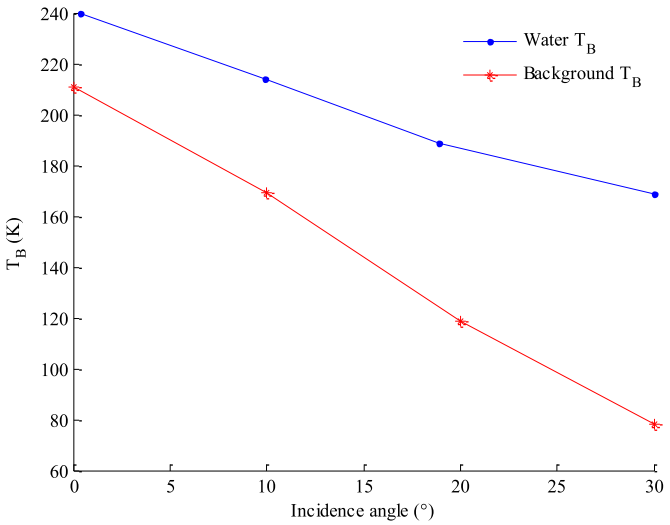


Fig. 9. Changing curves of T_B of water body and background with incidence angles under v polarization.

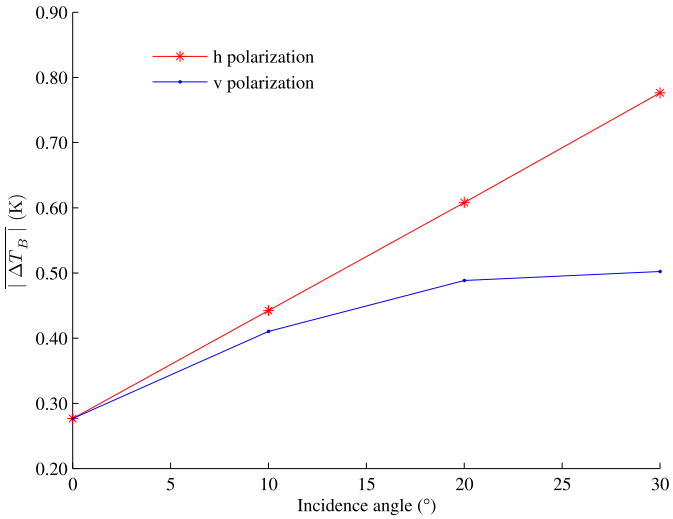


Fig. 10. $|\Delta \bar{T}_B|$-incidence angle curves by considering the variation of background T_B with incidence angle.

where T_{sur} refers to the background T_B. According to (26), the variation of $|\Delta T_B|$ with θ (referring to Fig. 6) is determined by the variation of $|\Delta e|$ with θ and the variation of T_{sur} with θ. In Fig. 7, the basic assumption is that T_{sur} does not change with θ, and it primarily reflects the variation of $|\Delta e|$ with θ. Therefore, the difference between Figs. 6 and 7 is mainly due to the influence of the variation of T_{sur} with θ. In (26), when θ increases, T_{sur} decreases; therefore, $[T_{obj} - T_{sur}(\theta)]$'s value goes up. However, in (26), the variation trend of $|\Delta e|$ is not only associated with θ but also polarization modes.

In the case of h polarization, $|\Delta e_h|$ increases with the increase of θ (see the red curve in Fig. 7). According to the monotonicity of the composite function, the composite function $|\Delta T_B|$, which is the product of two positive and increasing functions, $|\Delta e(\theta)|$ and $[T_{obj} - T_{sur}(\theta)]$, is still an increasing function. Therefore, $|\Delta T_B|$ also rises with the increase of θ under h polarization, and the increasing trend of the red curves in Figs. 6 and 7 is similar. However, since both $|\Delta e(\theta)|$ and $[T_{obj} - T_{sur}(\theta)]$ rise with θ, the two forward actions add up to make the variation rate (slope) of $|\Delta T_B|$ with θ be higher. In the case of h polarization, the calculation results show that the average increasing rate is 0.0121 K/° (0.0037 K/°) in Fig. 6 (Fig. 7). Hence, the former is greater than the latter, which is consistent with the above analysis results.

In the case of v polarization, $|\Delta e_v|$ decreases with the increase of θ (see the blue curve in Fig. 7). According to the monotonicity of the composite function, the composite function $|\Delta T_B(\theta)|$, which is the product of a decreasing function $|\Delta e_v|$ and an increasing function $[T_{obj} - T_{sur}(\theta)]$, may be an increasing or decreasing function. And it is determined by magnitude of $|\Delta e_v|$ and $[T_{obj} - T_{sur}(\theta)]$. Therefore, based on theoretical simulation, the $\overline{|\Delta T_B|} - \theta$ curves under h and v polarizations with considering the decrease of background T_B with θ were given in Fig. 10. In Fig. 10, the change trend of $\overline{|\Delta T_B|}$ with θ under v polarization is contrary to that in Fig. 7, and is consistent with that in Fig. 6. It means that, in the near-field conditions, the $\overline{|\Delta T_B|} - \theta$ variation trends in the samples under v polarization are determined by the variation of background radiation with θ.

In addition, according to (4), background radiation changes do not influence the trends of ΔT_B. T_{obj} is always greater than T_{sur} under the outdoor conditions. Moreover, the trends of ΔT_B are determined by Δe which is related to the changes of mineral dielectric constant. Therefore, the experimental results are reliable even under the near-field condition.

$$[\varepsilon]_a = \begin{bmatrix} \varepsilon_{xx} + \Delta\varepsilon_{ix}(\varphi) - \Delta\varepsilon_{ex}(\varphi) & 0 & 0 \\ 0 & \varepsilon_{yy} + \Delta\varepsilon_{iy}(\varphi) - \Delta\varepsilon_{ey}(\varphi) & 0 \\ 0 & 0 & \varepsilon_{zz} + \Delta\varepsilon_{tz}(\varphi) - \Delta\varepsilon_{ez}(\varphi) \end{bmatrix} \tag{21}$$

$$\varepsilon_a = \sqrt{[\varepsilon_{xx} - \Delta F(\varphi)]^2 \cos^2\alpha + [\varepsilon_{yy} - \Delta G(\varphi)]^2 \cos^2\beta + [\varepsilon_{zz} - \Delta H(\varphi)]^2 \cos^2\gamma} \tag{23}$$

$$\begin{aligned} \Delta\varepsilon = \varepsilon_a - \varepsilon_b &= \frac{\varepsilon_a^2 - \varepsilon_b^2}{\varepsilon_a + \varepsilon_b} \\ &= \frac{[\Delta F(\varphi)^2 - 2\varepsilon_{xx}\Delta F(\varphi)]\cos^2\alpha + [\Delta G(\varphi)^2 - 2\varepsilon_{yy}\Delta G(\varphi)]\cos^2\beta + [\Delta H(\varphi)^2 - 2\varepsilon_{zz}\Delta H(\varphi)]\cos^2\gamma}{\varepsilon_a + \varepsilon_b} \end{aligned} \tag{24}$$

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B. Sensitivity Analysis of T_B to the Change of Permittivity

The experimental and theoretical study on the anisotropy and polarization of the change of samples' T_B in the range of θ at 0°–30° and near-field conditions are discussed above. However, the satellite-based observations of the Earth are mostly collected with the incidence angles between the range of 40°–60°, belonging to far-field condition. T_B of the cold sky background can be considered unchanged with the incidence angles. Moreover, the stress-induced variation of permittivity, rather than temperature, is the main reason for the changes of T_B in granite samples [29], [30]. The unanswered question is to determine under which polarization mode and incidence angle, T_B of rocks is most sensitive to the stress-induced variation of permittivity. When the change of permittivity in all directions is kept the same, the larger the change of T_B caused the change of permittivity under the incidence angle is, the more sensitive T_B is to the change of permittivity. This makes it easier to detect larger microwave signals under the sensitive incidence angle. The same is true for the sensitive polarization. Therefore, the sensitivity analysis of T_B to the change of permittivity under θ at 0°–90° and far-field conditions are discussed below.

The permittivity of rocks decreases gradually from ultrabasic rocks, basic rocks, intermediate rocks, acid rocks to sedimentary rocks, ε' with a range of 9–5, and ε'' from 0.1 to 0.5 [44]. Thus, the average values of $\varepsilon' = 7$ and $\varepsilon'' = 0.3$ were taken for example rock here. According to the results of dielectric test on quartz pressed [29], the change of rock's permittivity under stress in each direction is also assumed to be -0.08. $\overline{|\Delta T_B|}$ of rock under different θ at 0°–90° and h and v polarizations were calculated based on (9) and (10), as shown in Fig. 11. From Fig. 11, it can be seen as follows.

1) Under h polarization, $\overline{|\Delta T_B|}$ of rock first increases and then decreases with θ, and reaches the maximum at θ_1 (52°). Under v polarization, it first decreases with θ from 0° to θ_2 (69°), then increases from θ_2 to θ_4 (84°), and finally decreases from θ_4 to 90°.
2) From 0° to θ_3 (80.5°), $\overline{|\Delta T_B|}$ of rock under h polarization is higher than that at v. Then, it is lower under h polarization from θ_3 to 90°. By comprehensive comparison, $\overline{|\Delta T_B|}$ reaches the maximum at h polarization and θ_1. This means that under the mentioned conditions, T_B of rock is most sensitive to the permittivity change.
3) It is noted that, under the conditions of v polarization and θ_2, $\overline{|\Delta T_B|}$ of the rock is close to 0. This means that T_B is least sensitive to the change of permittivity under these conditions. It is important because the changes of T_B under v polarization and θ_2 are subject mainly to temperature change, which is helpful in extracting the temperature information associated with EQs.

C. Potential Utility Using Satellite Microwave for EQ Study

1) Enlightenment for Extracting Microwave Anomalies Associated With EQs: The trends of the samples' ΔT_B have certain randomness, either uptrend, downtrend, or no obvious change. The variation trends of T_B with stress in the rock under investigation are of significance. It suggests that during the process of ground stress adjusting quickly in the seismic occurrence areas, the ΔT_B anomalies related to EQs may appear. Therefore, abnormal rise or fall of T_B deserves special attention since they may well be associated with EQ.

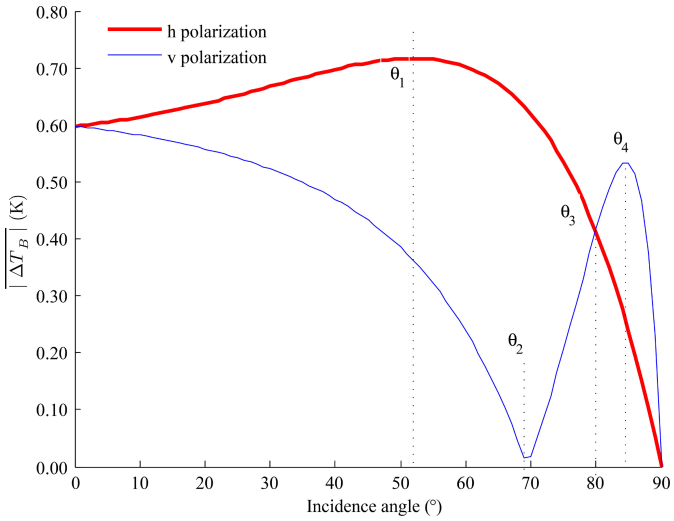


Fig. 11. $\overline{|\Delta T_B|}$–θ curves at 0°–90° and h and v polarizations. To facilitate the comparison of $\overline{|\Delta T_B|}$ under different polarizations at the same θ, θ at 0°–90° were divided into five intervals by four special angles, namely, θ_1, θ_2, θ_3, and θ_4, which are related to the permittivity in targets. For example, rocks with the real (imaginary) part of permittivity 7 (0.3), θ_1 is 52°, θ_2 is 69°, θ_3 is 80.5°, and θ_4 is 84°, respectively. At $\theta_1(\theta_2)$, $\overline{|\Delta T_B|}$ at h (v) polarization reaches its maximum (minimum). When $\theta < \theta_3$ ($\theta > \theta_3$), $\overline{|\Delta T_B|}$ at h (v) polarization is always greater than that at v (h) polarization. When $\theta > \theta_4$, the variation trends of $\overline{|\Delta T_B|}$ at v polarization with θ would transform.

The sensitivity of T_B to the change of dielectric constant is related to the incidence angles and polarization modes. For rocks, there exist the most sensitive and least sensitive incidence angles and polarization modes. T_B is most sensitive under the angle θ_1 and h polarization, which can be used to detect and extract the anomalies in microwave emissivity associated with the EQs. T_B is least sensitive under the angle θ_2 and v polarization, which can be used to detect and extract the ground temperature information associated with the EQs.

For the same sample, the trends of the samples' ΔT_B may be opposite under different θ and p. The anisotropy and polarization of ΔT_B are the special properties of the anisotropic and heterogeneous materials, like rocks. These properties are the important marks that distinguish ground surface T_B changes caused by the changes of meteorological conditions (e.g., rainfall and air temperature) and EQs. This is because when the moisture content changes, the trends of ΔT_B in each direction may be the same. So is the air temperature. This article's result can provide an experimental basis for coordinating multisatellite T_B to extract microwave anomalies associated with the EQs.

2) Reference Significance for Launching Satellite Dedicated to Detect EQ-Related Microwave Anomalies: ΔT_B of the granite samples is anisotropic. The magnitude of ΔT_B varies at different θ for the same samples. Even at a certain θ, T_B does not change with the stress. However, T_B changes significantly with stress exerted at another angle. This reflects the richness of the information of microwave radiation from rocks loaded

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in the half-infinite space. Therefore, multiangle and multidirection observation is helpful in detecting significant microwave abnormal signals associated with the EQs.

In the 5G/6G era, multipurpose satellites integrated communication, navigation, and remote sensing have become one of the development trends of geospatial information technology [45]. For 6G communications, the launch of multiple satellites into space is required, which may provide potential satellite-based platforms for seismic remote sensing. The global microwave monitoring platform with multidirection, multiangle, multipolarization, and even multifrequency band conditions combined with the future 6G communications technology can prove beneficial for further research. This may be a sound research direction for future study in detecting microwave anomalies associated with the EQs. In this article, the anisotropy and polarization of ΔT_B of granite samples were studied experimentally and theoretically. It is worth noting that the most sensitive θ and p are strongly correlated with the dielectric constant of the ground objects. In addition, the terrain type and rainfall are other important factors affecting the dielectric constant. It is challenging to solve the problem of the optimal θ and p of future seismic satellites, and further research is needed.

VI. Conclusion

In summary, the main findings of this research are as follows.

1) It was found that three change modes of the granite samples' T_B appeared with the increase of stress under θ at 0°–30°. They are uptrend, fluctuation, and downtrend, with a probability of 30.36%, 35.71%, and 33.93%, respectively, and a maximum rate of 0.048 K/MPa. The change modes, related to the heterogeneity of the samples, suggest that the ΔT_B anomalies related to EQs may appear differently with the ground stress adjusting quickly in the seismic occurrence areas. Moreover, abnormal rise or fall of T_B deserves special attention since they may well be associated with coming EQs.
2) It was also found that the stress-induced T_B changes (ΔT_B) of the granite samples are anisotropic and polarized. It was also concluded that the magnitude of ΔT_B differs with θ and p, and the magnitude of ΔT_B at h polarization is also prone to greater changes than that at v polarization, with a probability of 67.86%. In addition, the trends of ΔT_B may also be reversed at different $\theta(p)$ with a probability of 57.14% (14.29%). The anisotropy and polarization of ΔT_B relates to the anisotropy and polarization of the stress-induced change of the samples' permittivity. This is a result from the combined action of piezoelectric effect, extension and contraction of ionic bonds, and turning-direction polarization. Due to that a number of anisotropic and heterogeneous solid materials, like rocks, exist in Earth crust, the anisotropy and polarization of ΔT_B in the process of EQ preparation may occur. This may be an important mark that distinguishes the changes of the ground surface T_B caused by the changes of meteorological conditions (e.g., rainfall and air temperature) and the EQs. It also provides an experimental basis for the analysis of microwave anomalies associated with the EQs based on multisatellite data.
3) The sensitivity of T_B to the change in permittivity was analyzed by the theoretical simulation in order to reveal θ and p at which rock T_B is most sensitive to the changes in stress. For example, rocks with the real and imaginary part of permittivity being 7 and 0.3, the most (least) sensitive θ is around 52° at h polarization (69° at v polarizations). This is meaningful in selecting proper satellite data to extract the abnormal signals of microwave emissivity or ground temperature associated with the EQs.
4) However, one deficiency of the experiments is that the incidence angles were confined within t 0°–30° due to the limitation of experimental devices. The anisotropy and polarization of the change of dielectric constant needs to be further verified by other experiments. In addition, because water is a key factor that affects the dielectric property of rock, it is important to understand the response of wet rock to stress change. Therefore, further research is still needed in this particular issue.

Acknowledgment

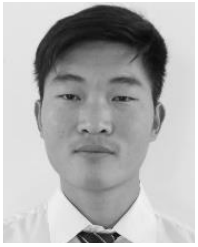
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References

[1] A. Tronin, "Satellite remote sensing in Seismology. A review," *Remote Sens.*, vol. 2, no. 1, pp. 124–150, Dec. 2009.
[2] Z. Qiang *et al.*, "Atellitic thermal infrared brightness temperature anomaly image—Short-term and impending earthquake precursors," *Sci. China Ser. D, Earth Sci.*, vol. 42, no. 3, pp. 313–324, Jun. 1999.
[3] V. Tramutoli, V. Cuomo, C. Filizzola, N. Pergola, and C. Pietrapertosa, "Assessing the potential of thermal infrared satellite surveys for monitoring seismically active areas: The case of kocaeli (°zmit) earthquake, august 17, 1999," *Remote Sens. Environ.*, vol. 96, nos. 3–4, pp. 409–426, Jun. 2005.
[4] M. Hashimoto *et al.*, "Crustal deformations associated with the great sumatra-andaman earthquake deduced from continuous GPS observation," *Earth, Planets Space*, vol. 58, no. 2, pp. 127–139, Feb. 2006.
[5] S. Stramondo, M. Moro, C. Tolomei, F. R. Cinti, and F. Doumaz, "InSAR surface displacement field and fault modelling for the 2003 bam earthquake (southeastern Iran)," *J. Geodynamics*, vol. 40, nos. 2–3, pp. 347–353, Sep. 2005.
[6] L.-X. Wu, K. Qin, and S.-J. Liu, "GEOSS-based thermal parameters analysis for earthquake anomaly recognition," *Proc. IEEE*, vol. 100, no. 10, pp. 2891–2907, Oct. 2012.
[7] J. Beçki, M. Parrot, and R. Wronowski, "Studies of the electromagnetic field variations in ELF frequency range registered by DEMETER over the sichuan region prior to the 12 may 2008 earthquake," *Int. J. Remote Sens.*, vol. 31, no. 13, pp. 3615–3629, Jul. 2010.
[8] T. Takano and T. Maeda, "Experiment and theoretical study of earthquake detection capability by means of microwave passive sensors on a satellite," *IEEE Geosci. Remote Sens. Lett.*, vol. 6, no. 1, pp. 107–111, Jan. 2009.
[9] T. Maeda and T. Takano, "Detection algorithm of earthquake-related rock failures from satellite-borne microwave radiometer data," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 4, pp. 1767–1776, Apr. 2010.

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[10] T. Maeda and T. Takano, "Detection of microwave signals associated with rock failures in an earthquake from satellite-borne microwave radiometer data," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Cape Town, South Africa, 2009, pp. 61–64.
[11] R. P. Singh, W. Mehdi, R. Gautam, J. Senthil Kumar, J. Zlotnicki, and M. Kafatos, "Precursory signals using satellite and ground data associated with the wenchuan earthquake of 12 may 2008," *Int. J. Remote Sens.*, vol. 31, no. 13, pp. 3341–3354, Jul. 2010.
[12] Y. Jie and G. Guangmeng, "Preliminary analysis of thermal anomalies before the 2010 baja california M7.2 earthquake," *Atmósfera*, vol. 26, no. 4, pp. 473–477, Oct. 2013.
[13] F. Jing, R. P. Singh, and X. Shen, "Land – atmosphere – meteorological coupling associated with the 2015 gorkha (M 7.8) and dolakha (M 7.3) nepal earthquakes," *Geomatics, Natural Hazards Risk*, vol. 10, no. 1, pp. 1267–1284, Jan. 2019.
[14] F. Jing and R. P. Singh, "Sensitivity of land covers on passive microwave brightness temperature," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Yokohama, Japan, Jul. 2019, pp. 9569–9572.
[15] F. Jing, R. P. Singh, Y. Cui, and K. Sun, "Microwave brightness temperature characteristics of three strong earthquakes in sichuan province, China," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 513–522, 2020.
[16] Y. Qi, L. X. Wu, M. He, and W. F. Mao, "Spatio-temporally weighted two-step method for retrieving seismic MBT anomaly: May 2008 Wenchuan earthquake sequence being a case," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 382–391, Jan. 2020.
[17] Y. Qi, L. Wu, W. Mao, Y. Ding, and M. He, "Discriminating possible causes of microwave brightness temperature positive anomalies related with may 2008 wenchuan earthquake sequence," *IEEE Trans. Geosci. Remote Sens.*, early access, Jul. 1, 2020, doi: 10.1109/TGRS.2020.3004404.
[18] H. Chen and Y.-Q. Jin, "A preliminary detection of anomalous radiation of rock failures related with yushu earthquake by using satellite-borne microwave radiometers," *Remote Sens. Technol. Appl.*, vol. 25, no. 6, pp. 860–866, 2010.
[19] Y. Ma, S. Liu, L. Wu, and Z. Xu, "Two-step method to extract seismic microwave radiation anomaly: Case study of MS8.0 wenchuan Earthquake," *Earthq. Sci.*, vol. 24, no. 6, pp. 577–582, Dec. 2011.
[20] F. Jing, R. P. Singh, K. Sun, and X. Shen, "Passive microwave response associated with two main earthquakes in tibetan plateau, China," *Adv. Space Res.*, vol. 62, no. 7, pp. 1675–1689, Oct. 2018.
[21] M. D. Deng, Z. F. Fan, C. Y. Cui, Q. Q. Ji, and N. G. Geng, "The experimental study for earthquake prediction by passive microwave remote sensing," *J. Infr. Millim. W.*, vol. 14, no. 6, pp. 401–406, 1995.
[22] K.-I. Maki, T. Takano, E. Soma, K. Ishii, S. Yoshida, and M. Nakatani, "An experimental study of microwave emissions from compression failure of rocks," *Zisin*, vol. 58, no. 4, pp. 375–384, 2006.
[23] T. Takano, H. Ikeda, and T. Maeda, "Consideration of the mechanism of microwave emission due to material destruction," *J. Appl. Phys.*, vol. 108, no. 8, Oct. 2010, Art. no. 083722.
[24] Z. F. Fang *et al.*, "Application of passive microwave remote sensing technology to the seismic prediction and its physical mechanism," *Chin. J. Geophys.*, vol. 15, no. 4, pp. 464–470, 2000.
[25] Z. F. Fan, Z. F. Fang, and M. D. Deng, "Basic experimental research of the microwave remote sensing applied in rock and soil engineering," *Chin. J. Radio Sci.*, vol. 15, no. 4, pp. 410–414, 2000.
[26] M. D. Deng, J. Y. Yin, Z. F. Fang, and Z. F. Fan, "Experimental research on the application of passive microwave remote sensing to the stability monitoring of concrete engineering," *J. Infr. Millim.*, vol. 20, no. 2, pp. 102–106, 2001.
[27] E. Y. Wang, Y. G. Wang, Z. H. Li, X. F. Liu, and E. L. Zhao, "Experimental study on the microwave radiation precursor laws of loading coal in deformation and fracture process," *Chin. J. Geophys.*, vol. 54, no. 9, pp. 2429–2436, 2011.
[28] S. Liu, Z. Xu, J. Wei, J. Huang, and L. Wu, "Experimental study on microwave radiation from deforming and fracturing rock under loading outdoor," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 9, pp. 5578–5587, Sep. 2016.
[29] X. Gao, S. Liu, L. Wu, W. Mao, W. Li, and J. Huang, "The variation in microwave brightness temperature of granite pressed under weak background radiation," *IEEE Trans. Geosci. Remote Sens.*, early access, Jun. 10, 2020, doi: 10.1109/TGRS.2020.2994439.
[30] W. Mao *et al.*, "Additional microwave radiation from experimentally loaded granite covered with sand layers: Features and mechanisms," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 5008–5022, Jul. 2020.
[31] J. W. Teng, Y. Q. Zhang, X. M. Ruan, G. Z. Hu, and Y. F. Yan, "The seismic anisotropy of the crustal and mantle medium of the earth interior and its dynamic response," *Chin. J. Geophys.*, vol. 55, no. 11, pp. 3648–3670, 2012.
[32] V. V. Sterlyadkin, "Radiometric microwave field measurements of the complex dielectric constant of the water surface," *Adv. Space Res.*, vol. 62, no. 11, pp. 3162–3169, Dec. 2018.
[33] J. K. Xiao, "Dielectric properties of minerals and rocks and their application in remote sensing," *Remote Sens. Environ.*, vol. 3, no. 2, pp. 136–145, 1988.
[34] W. Mao, L. Wu, and Y. Qi, "Impact of compressive stress on microwave dielectric properties of feldspar specimen," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 2, pp. 1398–1408, Feb. 2020.
[35] M. Y. Chen, *Electromagnetic field and microwave technology*. Beijing, China: Higher Education, 1989, pp. 15–17.
[36] S.-E. Park and T. R. Shrout, "Ultrahigh strain and piezoelectric behavior in relaxor based ferroelectric single crystals," *J. Appl. Phys.*, vol. 82, no. 4, pp. 1804–1811, Aug. 1997.
[37] R. J. Greene and E. A. Patterson, "An integrated approach to the separation of principal surface stresses using combined thermo-photo-elasticity," *Experim. Mech.*, vol. 46, no. 1, pp. 19–29, Feb. 2006.
[38] J. K. Xiao, Z. S. Lai, and J. M. Feng, "Microwave dielectric properties of rocks and soils and their applications in microwave remote sensing," *Chin. Sci. Bull.*, vol. 29, no. 8, p. 1123, 1984.
[39] J. K. Xiao, "The effect of mineral composition and structure on dielectric constant," *Acta Miner. Sinica*, vol. 5, no. 4, pp. 331–337, 1985.
[40] V. S. Merkulov, "Some aspects of reflection crystal optics," *Crystallogr. Rep.*, vol. 55, no. 2, pp. 258–261, Mar. 2010.
[41] G. G. Yang, and F. J. Song, *Advanced Physical Optics*. 2nd ed. Hefei, China: University of Science and Technology of China Press, 2008, pp. 229–230.
[42] J. L. Zhao, *Advanced Optics*. Beijing, China: National Defence Industry Press, 2002, pp. 83–85.
[43] S. M. Mikki and Y. M. M. Antar, "A theory of antenna electromagnetic near field—Part II," *IEEE Trans. Antennas Propag.*, vol. 59, no. 12, pp. 4706–4724, Dec. 2011.
[44] X. Wang, H. Guo, C. Wang, P. Chen, and Y. Shi, "Relative dielectric constant from dry rocks," *Chin. Sci. Bull.*, vol. 44, no. 24, pp. 2286–2293, Dec. 1999.
[45] D. R. Li, "Towards geospatial information technology in 5G/6G era," *Acta Geodaetica Cartographica Sinica*, vol. 48, no. 12, pp. 1475–1481, 2019.



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